SURNAME, FIRST NAME: EBU6230 Class Test 2, 2020/21

BUPT ID: **QM ID:** TOTAL MARK (examiner's use)

Q1. Which of the following masks is/are based on first-order derivative?

A: Sobel **B:** Roberts **C:** Laplacian **D:** Marr-Hildreth **E:** Canny

Q2. *True or False:* Sobel mask is more robust to noise than Prewitt mask.

A: True **B:** False

Q3. Harris corner detector is robust to

A: Image scale **B:** Rotation **C:** Noise **D:** Translation

Q4. Choose the correct statements about Moravec operator?

A: The process requires high computational load.
B: Diagonal edges can be detected as corners.
C: To capture the pixels with low contrast.
D: Can be applied in scale-space.

Q5. In Harris corner detector, a point is determined to be the corner based on the eigenvalues of the Harris matrix, λ_1, λ_2, when:

A: λ_1, λ_2 are large **B:** $\lambda_1 > \lambda_2$ **C:** $\lambda_1 < \lambda_2$ **D:** λ_1, λ_2 are small

Q6. In Morphological processing, choose the correct operation(s) that can perform boundary extraction. (I: image, s: structuring element, $\oplus$: dilation, $\ominus$: erosion, $\circ$: opening, $\bullet$: closing)

A: I$\circ$s **B:** s$\bullet$I **C:** (I$\oplus$S)-I **D:** (I$\ominus$s)-I **E:** I-(I$\ominus$s)

Q7. With dilation, the support of an image becomes

A: Thinner **B:** Thicker **C:** Shrinked **D:** Sharpened

Q8. Which of the following colour models are sued for colour printing?

A: RGB **B:** CMY **C:** CMYK **D:** HSV

Q9. Which colour model will you choose to apply Histogram equalization?

A: RGB **B:** CMY **C:** CMYK **D:** HSV

Q10. The colour transformation is processed between the

A: Dual colour model **B:** Single colour model **C:** Tri colour model **D:** All the above

Q11. Digitizing the image intensity amplitude is called?

A: Sampling **B:** Quantization **C:** Framing **D:** Both A and B

Q12. Coding redundancy works on

A: Pixels **B:** Matrix **C:** Intensity **D:** Coordinates

WRITE YOUR ANSWER UNDER EACH QUESTION:

Q13. Apply the Moravec operator to the centre point in the image and i) find the Moravec image value. Also, ii) determine if the point is corner or not. (Set the threshold for the corner to 15) **[4 marks]**

0	0	0	10	0
0	0	0	10	0
0	0	10	0	0
0	10	0	0	0
0	10	0	0	0

ANSWER

	20	

[2 marks]

Corner **[2 marks]**

Q14. Apply the erosion on the image *I* using the structuring element *s*, $I' = I \ominus s$. Note that the origin of s is coloured in grey. **[2 marks]**

$I =$

0	0	0	0	0
0	0	1	0	0
0	1	1	1	0
0	0	1	0	0
0	0	0	0	0

$s =$

0	1	0
1	1	1
0	1	0

ANSWER

$I' =$

0	0	0	0	0
0	0	0	0	0
0	0	0	1	0
0	0	0	0	0
0	0	0	0	0

[2 marks]

Q15.

a) Given a RGB image, c(x,y)=[r(x,y) g(x,y) b(x,y)], where each channel is quantized to 8 bit 0 to 255. What is the value of c(x,y) at the center point after apply image smoothing using a 3 x 3 average (mean) filter? **[2 marks]**

200	200	200
200	100	200
210	200	200

R

5	5	5
5	5	5
5	5	5

G

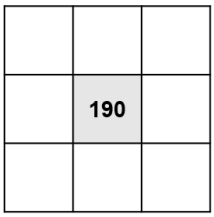
0	20	20
100	0	20
100	100	0

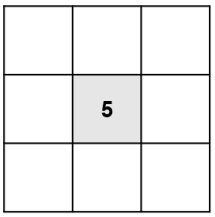
B

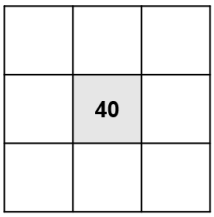
1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

3x3 average filter

Answer:







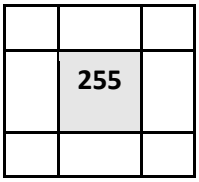
0.5 mark for each, if all are correct, 2 marks.

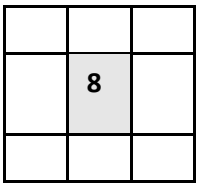
b) what is the output of the RGB values after applying the following transformation function:

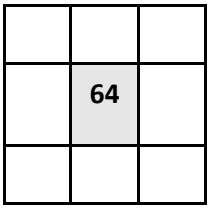
$$c'(x, y) = 1.6 \cdot c(x, y)$$

[3 marks]

Answer:







One mark for each

Q16. Describe the features of Region of Interest (RoI) coding principle**.**

[4 marks]

Answer:

Allows a non-uniform distribution of quality. The ROI is coded with a higher quality than the background (BG) [2 marks]

- A higher compression ratio can be achieved with same or higher quality inside ROIs**.** [2 marks]